

# Discrete Probability Review

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# Probability Space

A **probability space** is a triple  $(\Omega, \mathcal{F}, \Pr)$ :

- ▶  $\Omega$ : **Sample space** – the set of all possible outcomes.
- ▶  $\mathcal{F} \subseteq 2^\Omega$ : **Event space**. An event is a subset of the outcomes.
- ▶  $\Pr : \mathcal{F} \rightarrow [0, 1]$ : **Probability measure** – assigns a probability to each event.

Axioms (Kolmogorov):

1.  $\Pr(\Omega) = 1$ .
2.  $\Pr(A) \geq 0$  for all  $A \in \mathcal{F}$ .
3. If  $A_1, A_2, \dots$  are pairwise disjoint, then  $\Pr(\bigcup_i A_i) = \sum_i \Pr(A_i)$ .

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## Probability Space – Example

**Example:** Roll a fair six-sided die.

- ▶  $\Omega = \{1, 2, 3, 4, 5, 6\}$
- ▶  $\Pr(\{\omega\}) = 1/6$  for each  $\omega \in \Omega$
- ▶ The event “roll and odd number” would be  $\{1, 3, 5\}$

**Example:** Flip two fair coins.

- ▶  $\Omega = \{HH, HT, TH, TT\}$
- ▶  $\Pr(\{\omega\}) = 1/4$  for each  $\omega \in \Omega$
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# Events

An **event** is a subset  $A \subseteq \Omega$ .

- ▶ **A occurs** if the outcome  $\omega \in A$ .
- ▶ **Complement:**  $\bar{A} = \Omega \setminus A$ ,  $\Pr(\bar{A}) = 1 - \Pr(A)$ .
- ▶ **Union:**  $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$ .
- ▶ **Union bound:**  $\Pr(A \cup B) \leq \Pr(A) + \Pr(B)$ .

**Example:** Roll a die. Let  $A = \{2, 4, 6\}$  (even),  $B = \{1, 2, 3\}$  (at most 3).

- ▶  $\Pr(A) = 3/6 = 1/2$
- ▶  $\Pr(B) = 3/6 = 1/2$
- ▶  $A \cap B = \{2\}$ , so  $\Pr(A \cap B) = 1/6$
- ▶  $\Pr(A \cup B) = 1/2 + 1/2 - 1/6 = 5/6$

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# Independence

Events  $A$  and  $B$  are **independent** if

$$\Pr(A \cap B) = \Pr(A) \cdot \Pr(B).$$

**Example:** Flip two fair coins. Let  $A$  = “first coin is H”,  $B$  = “second coin is H”.

$$\triangleright \Pr(A) = 1/2, \quad \Pr(B) = 1/2$$

$$\triangleright A \cap B = \{HH\}, \quad \Pr(A \cap B) = 1/4 = \Pr(A) \cdot \Pr(B)$$

**Conditional probability:**

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \quad \text{provided } \Pr(B) > 0.$$

If  $A$  and  $B$  are independent, then  $\Pr(A \mid B) = \Pr(A)$ .



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# Random Variables

A **random variable**  $X$  is a function  $X : \Omega \rightarrow \mathbb{R}$ .

- ▶ Assigns a numerical value to each outcome.
- ▶ The **range** of  $X$  is the set of values  $X$  can take.

**Example:** Roll two dice. Let  $X = \text{sum of the two dice}$ .

- ▶  $\Omega = \{(i, j) : 1 \leq i, j \leq 6\}, \quad |\Omega| = 36$
- ▶  $X((i, j)) = i + j$
- ▶ Range of  $X$ :  $\{2, 3, \dots, 12\}$
- ▶  $\Pr(X = 7) = 6/36 = 1/6$

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# Probability Mass Function

For a discrete random variable  $X$ , the **probability mass function (PMF)** is

$$p_X(x) = \Pr(X = x).$$

Properties:

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## Expectation

The **expectation** (or expected value) of a discrete random variable  $X$ :

$$\mathbb{E}[X] = \sum_x x \cdot \Pr(X = x).$$

**Example:** Roll a fair die.  $X$  = value shown.

$$\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$$

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# Linearity of Expectation

For any random variables  $X_1, X_2, \dots, X_n$  (even if dependent!):

$$\mathbb{E} \left[ \sum_{i=1}^n X_i \right] = \sum_{i=1}^n \mathbb{E}[X_i]$$

**Example:** Expected number of heads in  $n$  coin flips.

- ▶ Let  $X_i = 1$  if flip  $i$  is heads, 0 otherwise.
- ▶  $\mathbb{E}[X_i] = p$  for each  $i$ .
- ▶  $\mathbb{E}[\sum_{i=1}^n X_i] = np$ .

For a constant  $c$ :  $\mathbb{E}[cX] = c \mathbb{E}[X]$  and  $\mathbb{E}[X + c] = \mathbb{E}[X] + c$ .

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## Linearity of Expectation

Puzzle:  $n$  students in a class. Each student randomly picks a name from a hat (with replacement). What is the expected number of students who pick their own name?

- ▶ Let  $X_i = 1$  if student  $i$  picks their own name, 0 otherwise.
- ▶  $\mathbb{E}[X_i] = 1/n$  for each  $i$ .
- ▶  $\mathbb{E}[\sum_{i=1}^n X_i] = n \cdot (1/n) = 1$ .

## Markov's Inequality

If  $X \geq 0$ , then for any  $a > 0$ :

$$\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

**Example:** Average grade is 70. What is the probability that a random student scores  $\geq 90$ ?

$$\Pr(X \geq 90) \leq \frac{70}{90} = \frac{7}{9} \approx 0.78$$

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